ME 264 Mechanisms Synthesis & Analysis I

Unit 3&4 Position, Velocity and Acceleration Analysis

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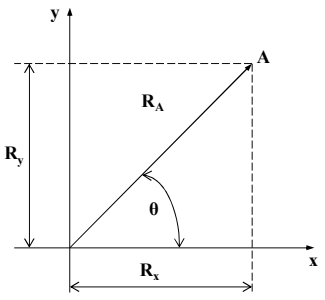
Introduction

- A tentative mechanism design may be analysed graphically or analytically to determine displacement, velocity and acceleration of moving parts.
- Position analysis is required for velocity, acceleration and forces analysis.
- Complete motion analysis includes virtually moving the linkage through its intended motions, which requires examination of the linkage at various positions.
- Graphical method is convenient for position analysis up to three positions. However, for multiple positions analysis, analytical method is more convenient and accurate than the graphical method.

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Position and Displacement Analysis

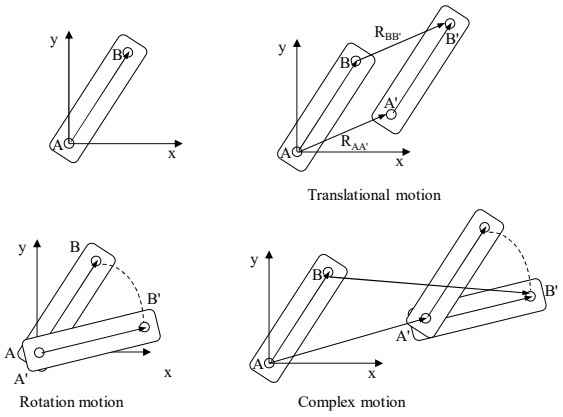
- A position of point in a plane may be defined as the position vector of the point from a given reference axes (or a point).
- The choice of the reference axes or point is arbitrary or/and may be selected to suit the observer.
- A 2D position vector has two attributes, which may be expressed in *polar* or *Cartesian* coordinates.
- In the polar form, the position vector is expressed in terms of its magnitude and angle. The Cartesian form is expressed in the x and y components of the vector.



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Rigid Body Motion

- Rigid bodies can exit three types of motion: translation, rotation and complex motion.
- In **translational** motion, a body moves such that all points on the body have the same displacement (or parallel).
- Translation along a straight path is called rectilinear translation (or motion), and along a curved path is curvilinear translation (or motion).
- In **rotational** motion, a body moves such that one point remains at the origin and all other points move with the same angular displacement.
- Complex motion is a combination of translational and rotation motions



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Graphical Position/Displacement Analysis of Linkage

- Graphical displacement analysis is trivial exercise, but very tedious as each link's position must redone.
- The algebraic or vector approach is less tedious as one expression is required for all positions, but it is complex and difficult.
- Displacement analysis of a single DoF mechanism such as a four-bar mechanism may be performed graphically by generating a number of positions of the mechanism.

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Graphical Position/Displacement Analysis of Linkage

- A quick method for generating a number of positions of a mechanism, as follows:

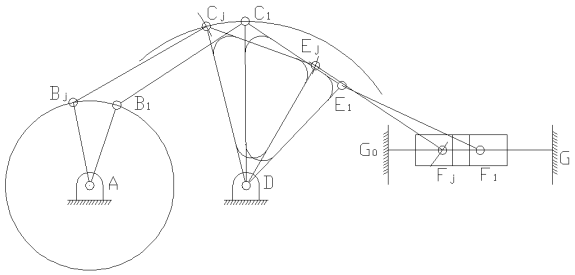
1. Construct the starting or initial position of the mechanism and marked each joint with the subscript.
2. Draw all known paths of joints. A revolute (pin jointed point) has a circular/arc path and prismatic (slider) joint has curve or straight path along the sliding path.
3. Always start from known position/path to unknown position/path.
4. For each point two intersecting paths are need for the next position.
5. Linear distance from one position to another gives linear displacement.
6. Angular displacement is the angle from one position to another

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Example 3:1

Determine position j of the linkage and find all displacements, if crank AB is input and B_j is given

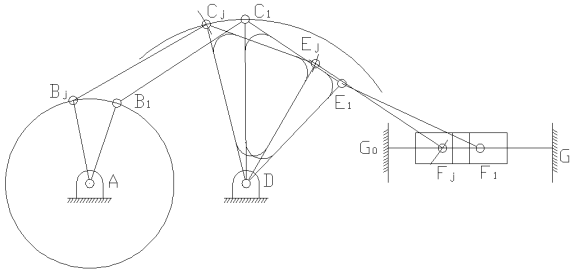
1. Draw to scale the linkage at position 1.
2. Place compass at A, extend to B1 and draw a circular path of B. Mark B_j on the path, knowing the angular displacement from B_1 to B_j.
3. Place compass at D and extend to C1 and draw a circular path of C. With the radius B_1C_1, place compass at Bj and describe arc to intersect the path C from D. Mark the intersection point as point C_j.



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Example 3:1 Determine position j of the linkage and find all displacements, if crank AB is input and B_j is given

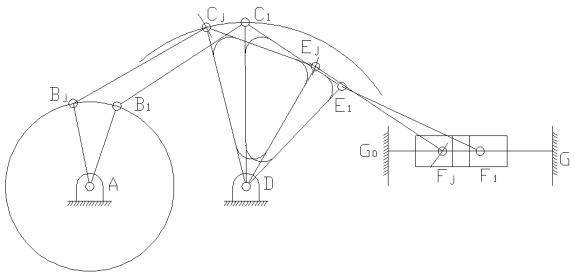
4. Place compass at D and extend to E_1 and draw a circular path of E. With the radius C_1E_1, place compass at Cj and describe arc to intersect the path E from D. Mark the intersection point as point E_j.
5. Draw a straight line through G_0G to mark the path of slider. With the radius E_1F_1, place compass at Ej and describe arc to intersect slider path G_0G. Mark the intersection point as point F_j.



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Example 3:1 Determine position j of the linkage and find all displacements, if crank AB is input and B_j is given

6. Measure displacements
   i. Angle $<B_j A B_1$ = angular displacement of crank AB
   ii. Angle $<Ej D E1 = <Cj D C_1$ = angular displacement of rocker CDE
   iii. Distance F_jF_1 = linear displacement of slider
   iv. Angle between lines B_jC_j and B_1C_1 = angular displacement of coupler BC
   v. Angle between lines E_jF_j and E_{1F1} = angular displacement of coupler EF



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Example 3:2 (Q2 KNUST ME 264 Exams May 2018)

1. The linkage shown is a hydraulic press. The hydraulic actuator EFG is pivoted to the main frame at point F and joined to the link DCE at point E. Part EF of the hydraulic actuator is extensible whereas part FG has fixed length. Link DCE, which has three pivot attachment points, is pivoted to the main frame at D and connects to coupler BC at C. The Coupler BC is connected to a slider at B, which slide through the main frame. Suppose hydraulic actuator extends to cause the ram attached to the left end of the slider to move from point A_1 to point A_2.

   (a)Use the graphical method to determine position of the press mechanism. Label all the moving joints with subscript

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(b) From your construction in (a) above,

(i) determine the angular displacement of link DCE. Leave your answer in degrees, and state whether it is clockwise or anti-clockwise direction.

(ii) determine the angular displacement of link BC from its initial position. Leave your answer in degrees, and state whether it is clockwise or anti-clockwise direction.

(iii) determine the angular displacement of the hydraulic actuator EFG. Leave your answer in degrees, and state whether it is clockwise or anti-clockwise direction.

(iv) determine the actual extension of the hydraulic actuator EFG, if the actual displacement of A_1A_2 = 150 mm.

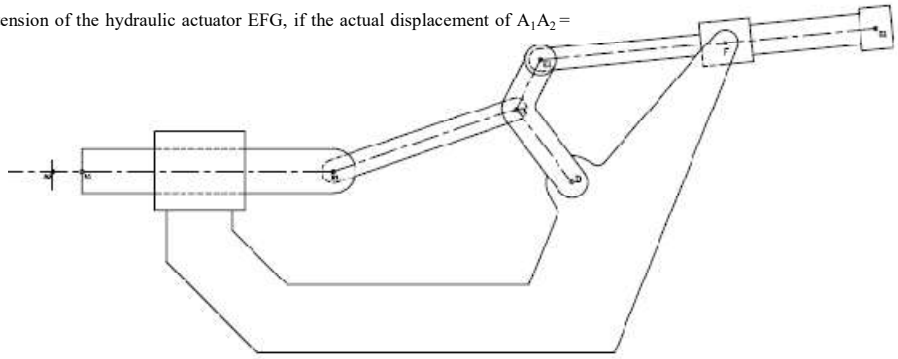


Figure 1

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Solution Steps

1. Draw a horizontal line through A1 and A2 and extend through B1 as path of slide AB.

2. Place compass at A_1, extend to B_1 to get radius A_1B_1, and place compass at B_2 and describe arc to intersect the sliding path of AB. Mark the intersection point as point B_2.

3. Place compass at D and extend to C_1 and draw a circular path of C. With the radius B_1C_1, place compass at B_2 and describe arc to intersect the path of C from D. Mark the intersection point as point C_2.

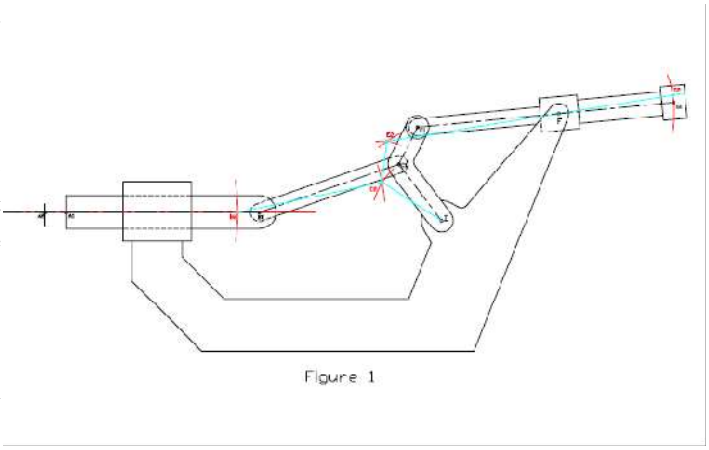


Figure 1

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Solution Steps

4. Place compass at D and extend to E_1 and draw a circular path of E. With the radius C_1E_1, place compass at C_2 and describe arc to intersect the path E from D. Mark the intersection point as point E_2.

5. Draw a line from E_2 to F and extend to pass point G_1. Place compass at F and extend to G_1 and draw a circular path of G to intersect extension of line E_2F. Mark the intersection point as point G_2

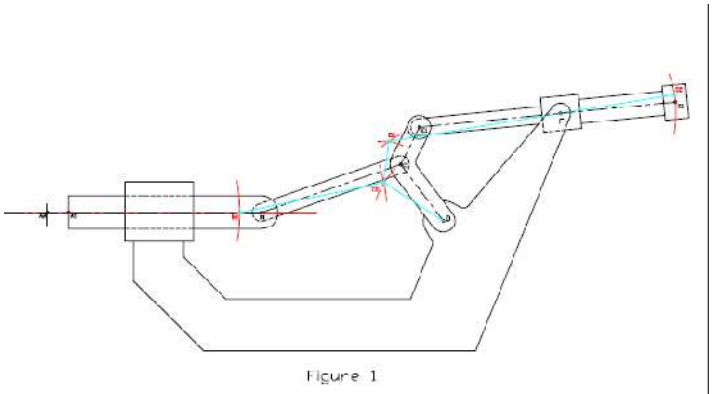


Figure 1

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Example 3:2 (Q2 KNUST ME 264 Exams May 2018)

(a)Measurements

(i) Angular displacement of DCE: Measure angle $<C_1DC_2 = 19.9^o$ anti-clockwise

(ii) Angular displacement of BC: Measure angle between B_1C_1 and $B_2C_2 = 7.2^o$ clockwise

(iii)Angular displacement of the hydraulic actuator EFG: Measure $<E_1FE_2$ or $<G_1FG_2 = 3.6^o$ anti-clockwise

(iv)determine the actual extension of the hydraulic actuator EFG: Measure extension E_1F and E_2F

Extension EFG= $E_2F - E_1F = 64-52 = 12$

Drawing $A_1A_2 = 8$ and Actual $A_1A_2 = 50$ mm

Therefore, actual extension of EFG = 12 x 150/8 = 225 mm

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Velocity Analysis

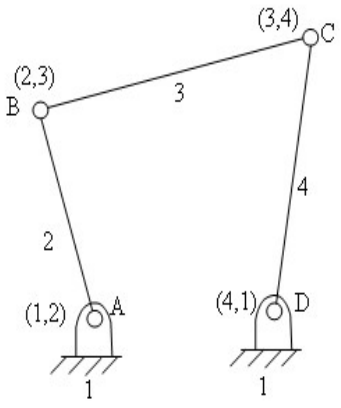
Three methods are used for velocity analysis

- Graphical methods
  - Instant Centre Method
  - Relative Velocity Method
- Analytical Methods
  - Vector Method
  - ~~Complex Method~~

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Instant Method

- An instant (or instantaneous) centre (IC) is a point where there is no relative velocity between two links of a mechanism at that instant.
- In any mechanism, obvious instant centres are joints.
  - In the four-link mechanism, there is no relative motion between link 1 and link 2 at joint A.
  - Similarly, there are no relative motions between links 2 and 3 at joint B, between links 3 and 4 at joint C, and between links 4 and 1 at joint D.
  - Therefore, the joints at A, B, C and D are instant centres.



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Number of Instant Centres

The number of instant centres of a linkage is determined by pairing all the links,

$$N = C_2^n = \frac{n(n-1)}{2}$$

where N = number of instant centres and n = number of links

Example 3:3

Find the number of instant centres of a fourbar mechanism

$$N = \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$$

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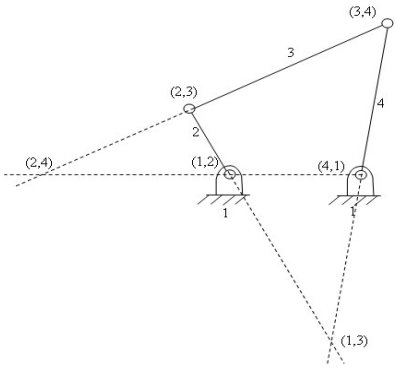
Kennedy's Theorem

- Kennedy's theorem is used to determine the instant centres of non-obvious instant centres.
- It deals with three instant centres of three links in a mechanism.
- Kennedy's theorem states that *three instant centres of three links moving relative to another must lie on a straight line.*

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Use of Kennedy's theorem to find IC

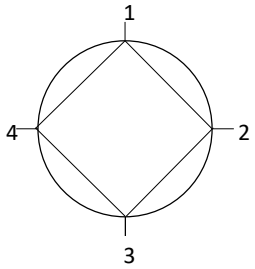
- The obvious ICs are (1, 2), (2,3), (3,4) and (4,1).
- ICs (2, 3) and (3, 4) must lie on the same line and are connected to the links 2, 3 and 4. Since the number 3 is common to centres (2, 3) and (3, 4), the left-out combination is (2, 4). Hence, the non-obvious centre on the line is (2, 4).
- Similarly, the non-obvious instant centre on the line connecting instant centres (1, 2) and (4, 1) is (2, 4).
- The point of intersection of lines through (2, 3) and (3, 4), and through (1, 2) and (4, 1) is the instant centre (2, 4).
- Similarly, the instant centre (1, 3) is the point of intersection of line through (3, 4) and (4, 1), and through (2, 3) and (1, 2).
- For complex mechanisms (i.e. more than four links), keeping track of instant centres already obtained and those that are obtainable is very helpful.



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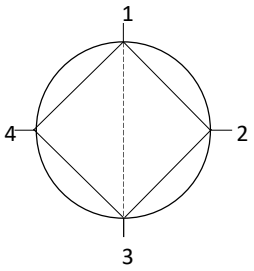
Use of Tracking Chart to Find Instant Centres

A tracking chart consists of a circle marked with ticks and numbers, each representing a link of the mechanism.



Steps for Tracking with Chart

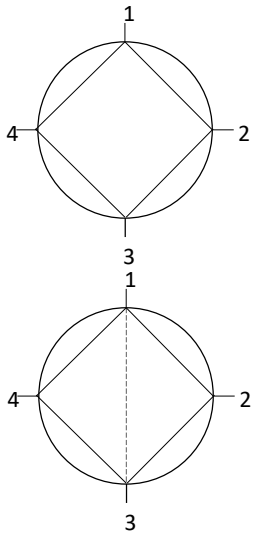
1. Draw a circle with any arbitrary radius
2. Mark on the circle with N number of ticks approximately equally spaced, where N = number of elements in the linkage.
3. Number the ticks from 1 to N, each representing an element.
4. Draw continuous lines from one tick to another to represent known instant centres. For example, the continuous line connecting 1 and 2 represents the instant centre (1, 2).



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Use of Tracking Chart to Find Instant Centres

A tracking chart consists of a circle marked with ticks and numbers, each representing a link of the mechanism.



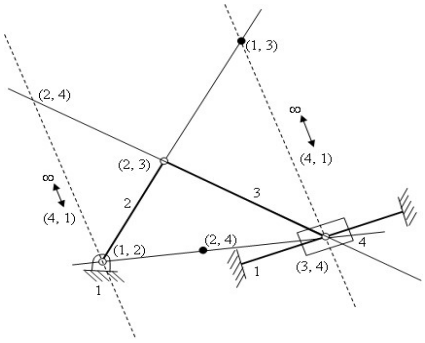
Steps for Tracking with Chart

5. To determine an unknown instant centre (I1, I2), check if you can get two independent triangles from the line (I1, I2).
6. If yes to the above, draw a dotted line from I1 to I2 and identify the two triangles.
7. For each triangle draw a line from one known IC to another on the figure.
8. The intersection of the two lines is the location of IC (I1, I2).
9. Label the IC (I1, I2) and change the dotted line (I1, I2) on the tracking chart immediately to continues line.
10. Repeat steps 5 to 9 until all unknown ICs are known.

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Instant Centre of Sliding Joint

- The instant centre of slider joint is not necessary at the point where the two links meet.
- It is off at infinity in the direction perpendicular to the sliding direction. Thus, the centre is at any point on any line perpendicular to the direction of sliding.
- Shown is a slider mechanism with sliding joint at (4, 1). In the figure, the instant centre (4, 1) lies on the two dotted lines.
- The instant (1, 3) is located at the point of intersection of the lines through (1, 2) and (2, 3), and through (4, 1) and perpendicular to sliding direction.
- Similarly, instant centre (2, 4) is identified by construction a lines through (2, 3) and (3, 4), and through (1, 2) and perpendicular to the sliding direction.



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$\vec{CB}$

Determination of Velocity from Instant Centre

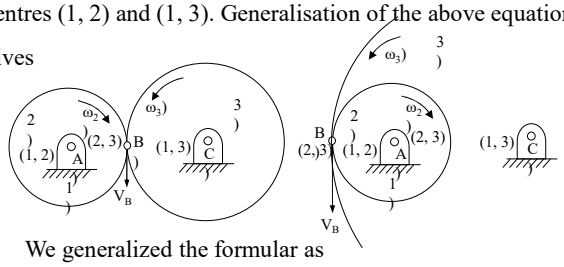
- Consider a pair of gears shown that produce the same instant angular velocity ratio as a pair of links
- From the figure, and given pitch radii of gears

$$V_B = \omega_2 \vec{AB} = \omega_3 \vec{CB}$$

Where V_B is the same for both gears and tangential velocity

$$\Rightarrow \frac{\omega_2}{\omega_3} = \frac{\vec{CB}}{\vec{AB}} = \frac{-(1,3-2,3)}{-(1,2-2,3)} \qquad \frac{\omega_2}{\omega_3} = \frac{(1,3-2,3)}{(1,2-2,3)}$$

From the figure, the equation is negative when the instant centre (2, 3) lies between the instant centres (1,2) and (1,3), and positive when the instant centre (2, 3) lies between the instant centres (1, 2) and (1, 3). Generalisation of the above equation gives



We generalized the formular as

$$\frac{\omega_n}{\omega_m} = \frac{(1,m-n,m)}{(1,n-n,m)}$$

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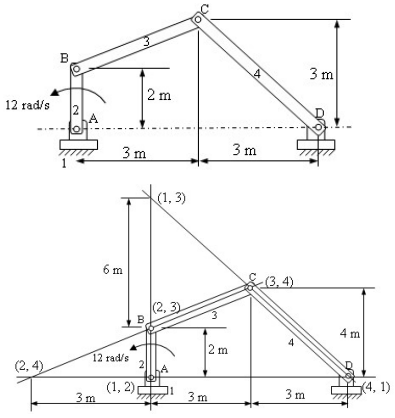
Example 3.4

If bar AB rotates at 12 rad/s in the anti-clockwise direction, determine the angular velocity of BC and CD with respect to the ground.

$$\frac{\omega_3}{\omega_2} = \frac{(1,2-2,3)}{(1,3-2,3)} = -\frac{2}{6} \qquad \omega_3 = -\frac{2}{6}(12) = -4$$

$$\frac{\omega_4}{\omega_2} = \frac{(1,2-2,4)}{(1,4-2,4)} = \frac{3}{9} \qquad \omega_{BC} = -4$$

$$\omega_3 = \frac{3}{9}(12) = 4 \qquad \omega_{CD} = 4$$

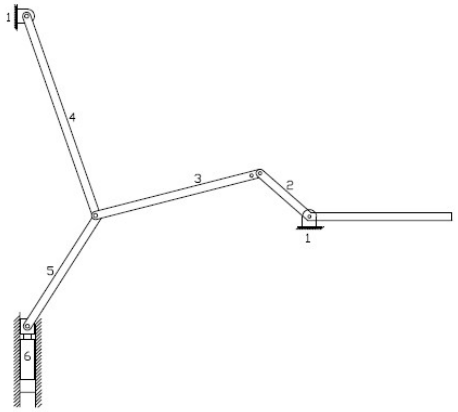


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Example 3.5

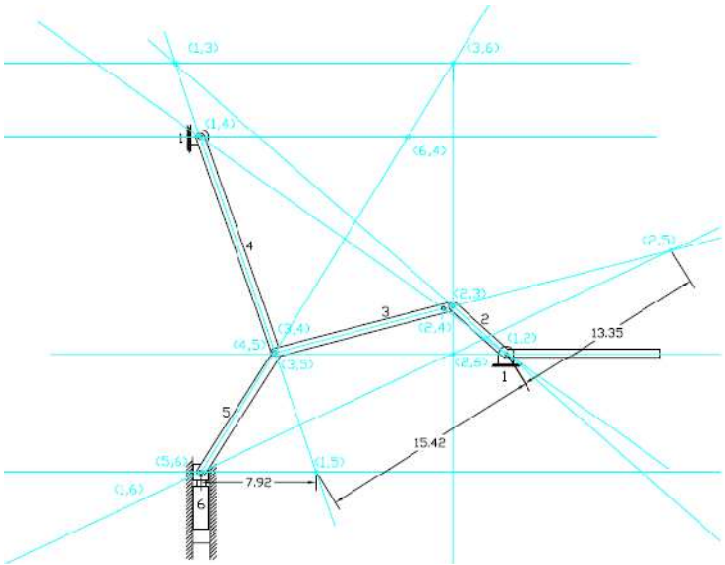
a) Locate all the instant centre of the linkage shown.

b) If Link 2 is rotating at 10 rad/s, determine the sliding velocity of the slider 6. All dimensions are in cm.



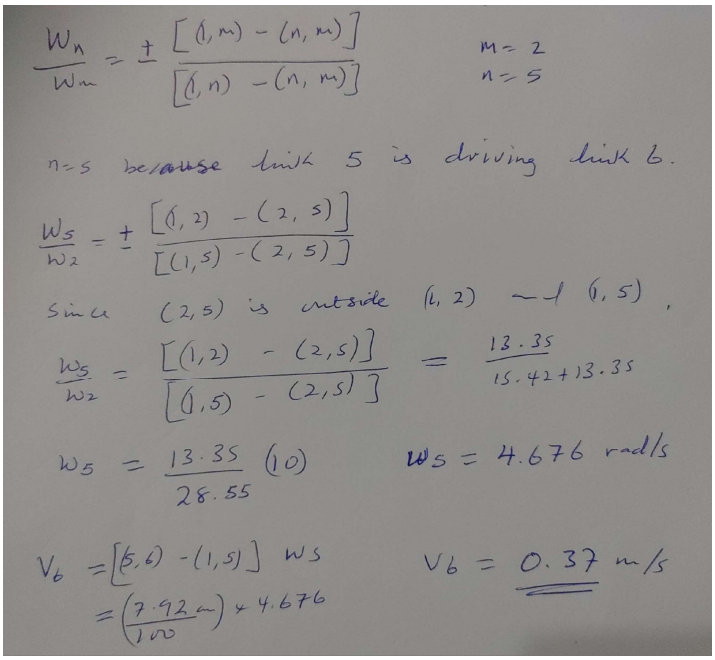
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(a) Locate all the instant centre of the linkage shown.



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b) If Link 2 is rotating at 10 rad/s, determine the sliding velocity of the slider 6. All dimensions are in cm.



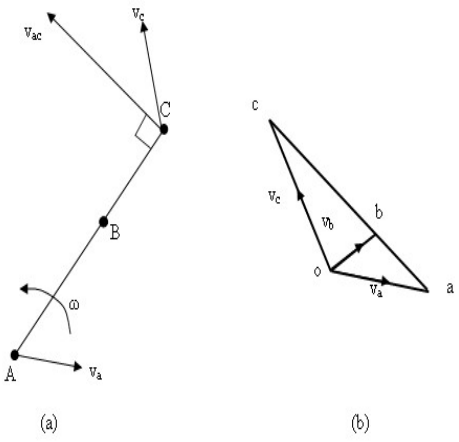
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Relative Velocity Method

- Consider the ends of a rigid link AC in moving with absolute velocities v_a and v_c, respectively.
- Note that these velocities are relative to the fixed point O. If the link is not extending, then, difference in velocity is due to only rotation about a point.
- Therefore, the link may be regarded as rotating about a point A with linear velocity given by

$$v_{ac} = \omega \mathrm{x} |AC| \qquad (1)$$

- where v_{ab} is the velocity of point C relative A and ω is angular velocity of the link AC.



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Relative Velocity Method

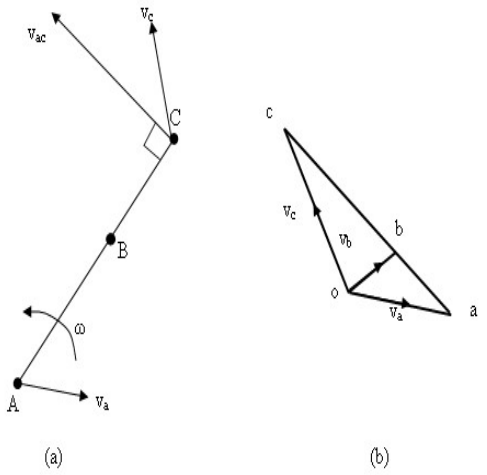
- The direction of v_{ac} is perpendicular to the line connecting points A and C because it is due to rotation.
- The velocity of point B, which is located on the line from A to C, is also perpendicular the line AC.

$$v_{ab} = \omega |AB| \qquad (2)$$

From (1) and (2), we eliminate ω and we have

$$v_{ab} = v_{ac} \frac{|AB|}{|AC|} \qquad (3)$$

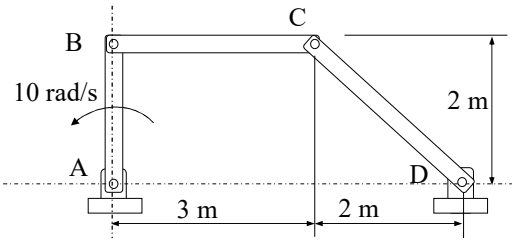
- The relative velocity of point B is located on the line ac of the velocity diagram using proportionality relationship.



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Example 3.6

If bar AB rotates at 10 rad/s in the anti-clockwise angular direction, determine the instantaneous angular velocity of bar CD.



Solution

The fixed (or zero-velocity) points are A and D, and are indicated with lower cases a and d, respectively. The velocity of point B is

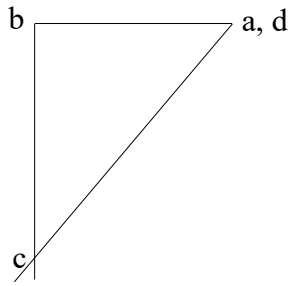
$$v_{ab} = \omega |AB| = 10x2 = 20 \text{ m/s}$$

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- Since AB is rotating in the anti-clockwise direction, vector ab points to the left and is perpendicular to AB.
- The velocity of C relative B is perpendicular to BC and the velocity of C relative D is perpendicular to CD. Hence, the intersection of lines through b and d, perpendicular to BC and DC respectively, yields the point c.
- Note that the points A and D are on the ground link and have zero velocities. Therefore, the two points have the same relative velocity of diagram.
- From the relative velocity diagram, velocity of point C relative d is 28.24 m/s.
- The velocity of point C is

$$v_C = w_{CD} \text{ x } r_{C/D} = w_{CD} \text{ x } |CD|$$

$$\Rightarrow \omega_{CD} = \frac{v_C}{|CD|} = \frac{28.28}{\sqrt{(2^2 + 2^2)}} = 10 \text{ rad/s}$$

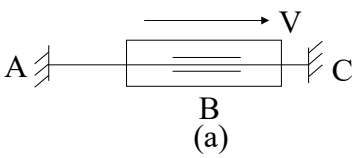


Relative Diagram

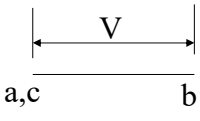
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Relative Velocity: General Motion with Sliding on Fixed Link

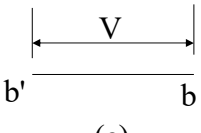
- Consider a rigid bar B sliding on a rigid link AC shown in (a)
- The velocity of B relative to the link AC is in the same direction as (or parallel to) the sliding direction AC, which is illustrated in (b).
- If B is moving towards the right, then b will be on the left of a, c. Note that A and C are fixed (zero-velocity) points, and therefore have zero velocities.
- Notice that there are two points at point B: one point on the slider and other on the sliding line AC. If the point on the sliding line AC and on the slider B are denoted B' and B, respectively, then the a, c may be replaced by b' as in shown (c).



(a)



(b)

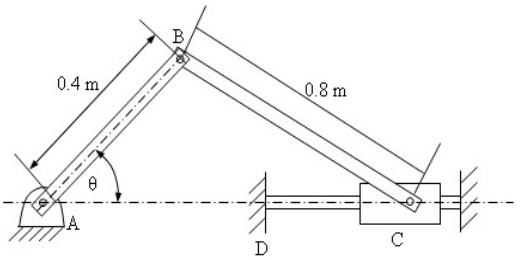


(c)

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Example 3.7

If angle θ = 45° and link AB has a constant anti-clockwise angular velocity of 5 rad/s, determine the sliding velocity of C.



Solution

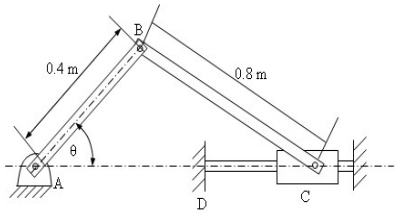
The velocity of point B is

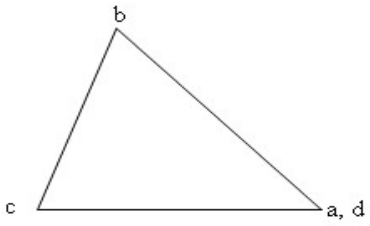
$$v_{ab} = \omega |AB| = 5x0.2 = 1.0 \text{ m/s}$$

33

Example 3.8

- Let point C' and C be points sliding line and slider C, respectively.
- Velocity of B relative A (line ab on velocity diagram) is perpendicular to link AB and points upward.
- The line ab is perpendicular to AB with length equivalent to 1.0 m/s.
- The velocity of C relative to D (C') is parallel to the sliding direction, which is given by line dc.
- To locate point c, draw a line through b and perpendicular to BC and another line through a, d and parallel the sliding direction. The intersection of two lines is point c. velocity diagram.
- From the relative velocity diagram, the velocity of the slider is dc, which is 0.974 m/s.





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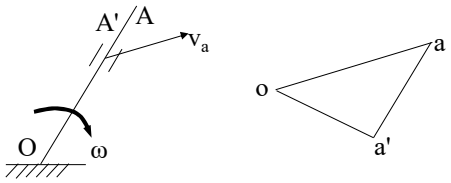
is parallel to OA

Relative Velocity: General Motion with Sliding on Rotating Link

- Consider link OA which rotating about point O and slider A sliding link OA with angular velocity ω.
- Let A' and A be the points on the rotating link and sliding link, respectively. The two points are coinciding at point A.
- The velocity of A' relative to point O is given is

$$v_A = \omega |OA|$$

  - Velocity V_A (denoted a' of relative diagram) is perpendicular to link OA and points downward.
  - Sliding velocity a relative a' is parallel the sliding direction

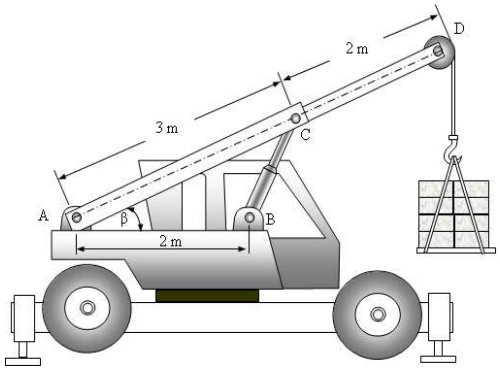


(a) (b)

35

Example 3.9

The hydraulic actuator BC of the crane is extending at a constant rate of 0.2 m/s. When the angle β = 35°, what is the angular velocity of the crane's boom AD.



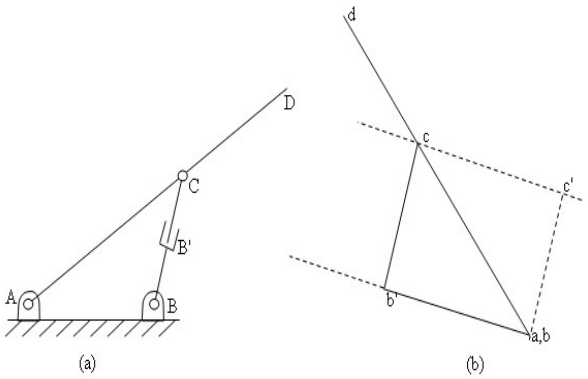
36

Solution

- Equivalent mechanism and velocity diagram are shown. Let B' be an imaginary point on the link BB'.
- First, the velocity of B' relative to B is drawn perpendicular to the line B- B' and on the left side of point (a, b) on the velocity diagram.
- The point b' on the relative velocity is on the left-hand side of b because as the actuator BC expands, BB' turns in the anti-clockwise direction.
- The velocity of C relative to A is drawn perpendicular to AC and on the left-hand side of C because AC also turns in the anti-clockwise direction.
- Along the direction of BC, the velocity of C relative B' is drawn through b to say point c' with dotted or construction lines.
- The length of b'c has an equivalent magnitude of 0.2 m/s. A line through c and parallel to bc' is draw to meet line ac. The point where the line through c' meets line ac is point c.

$v_C = 0.3104$ m/s

$$\omega_{AD} = \frac{v_C}{|AC|} = \frac{0.3104}{3} = 0.1035\,\text{rad/s}$$



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Example 3:10
ME 264 end of semester Exams (KNUST 2016)

1. The driving crank AB of the mechanism shown in Figure 5 is rotating at 10 rad/s in anticlockwise direction. The lengths of various links are: AB = 50 mm; BC = 350 mm; CD = 125 mm; BE = 200 mm; EG = 200 mm and GH = 300 mm. Other dimensions are shown on the figure. The swivel joint F is pinned to the ground and allows both sliding and rotation motions at the same time. Draw the relative velocity diagram for the given position. From the diagram, determine the [**13 points**]

(a) the angular velocity of link CD about pivot D. [**4 points**]

(b) the sliding velocity of slider H. [**2 points**]

(c) the angular velocity of link EG about the swivel joint F. [**4 points**]

(d) the sliding velocity of link EG through the swivel joint F. [**2 points**]

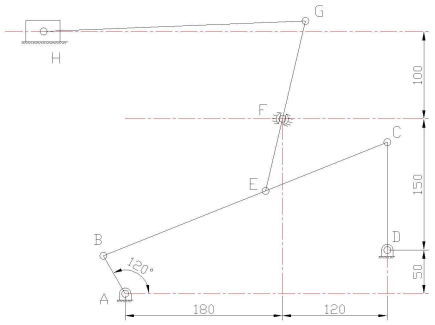
38

Example 3:10 Solution

- Let point H and H' be points on sliding line and slider H, respectively.
- Let point F and F' be points on ground at swivel joint F, and on link EFG, respectively.
- Velocity of B relative A (line ab on velocity diagram) is perpendicular to link AB and points upward. The line ab is perpendicular to AB with length equivalent to 10x50/1000=0. 50 m/s.
- To locate c, a lines perpendicular to BC and through b and perpendicular to DC and through are drawn. Intersection is c.
- Since E is on link CB, |eb|=|cb||EB|/|CB|. We measure cb, EB and CB; calculate eb and locate e. Note that E is between C and B; therefore e will be between b and c.
- We draw a line parallel to EG through f on velocity diagram to represent sliding velocity of EG through F (ground). Another line is drawn perpendicular to EG and through e. Intersection of the two lines gives f'.
- Since G is on link EFG, |eg|=|ef'||EG|/|EF|. We measure ef', EG and EF; calculate eg and locate g. Note the position of G relative to E and F, and locate g relative to e and f' with relativity.
- The velocity of H' relative to H (ground) is parallel to the sliding direction. Draw a line through h and parallel to the sliding direction; and another line perpendicular to GH and through g. Intersection of the two lines gives h'.

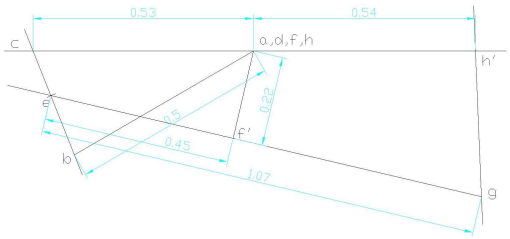
39

Example 3:10 Solution



All dimensions in mm

Figure 3



(a) the angular velocity of link CD , ω_{CD}= |cd|/|CD| =0.53/0.125

ω_{CD}= 4.24 rad/s

(b) the sliding velocity of slider H, V_H=hh' = 0.54 V_H= 0.54 m/s

(c) the angular velocity of link EG about the swivel joint F, ω_{FG}= |f'e|/|FE| =|f'g|/|FG| =0.62/0.1153 ω_{FG}= 5.38 rad/s

(d) the sliding velocity of link EG through the swivel joint F, V_F=ff' = 0.22

V_F= 0.22 m/s

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Velocity Analysis using the Vector Method

$$r_B = r_A + r_{B/A} \qquad (1)$$

- Where $r_{B/A}$ is the position of point B relative to point A. Let

$$r_{B/A} = xi + yj + zk \qquad (2)$$

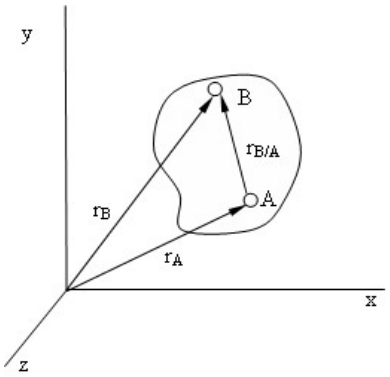
Differentiation Equation (1) with respect to time gives

$$v_B = v_A + \left(\frac{dx}{dt}i + x\frac{di}{dt}\right) + \left(\frac{dy}{dt}j + y\frac{dj}{dt}\right) + \left(\frac{dz}{dt}k + z\frac{dk}{dt}\right)$$

$$v_B = v_A + \left(\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right) + \left(x\frac{di}{dt} + y\frac{dj}{dt} + z\frac{dk}{dt}\right) \qquad (3)$$

Differentiating the unit vectors with respect to time only causes rotation of the vectors. Thus,

$$\frac{di}{dt} = w \times i \qquad \frac{dj}{dt} = w \times j \qquad \frac{dk}{dt} = w \times k \qquad (4)$$



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Velocity Analysis using the Vector Method

Substituting the above equation (4) into equation (3), we have

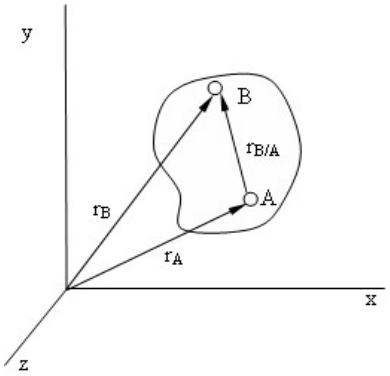
$$v_B = v_A + \left(\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right) + \left[x(\omega \times i) + y(\omega \times j) + z(\omega \times k)\right]$$

$$v_B = v_A + \left(\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right) + \omega \times \left[xi + yj + zk\right]$$

$$v_B = v_A + v_{B/A} + w \times r_{B/A} \qquad (5)$$

where $v_{B/A} = \left(\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right)$

v_A and v_B are respectively velocities of point A and B relative to the reference frame, and $V_{B/A}$ is the sliding velocity of point B relative to point A



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Velocity Analysis using the Vector Method Cases

- Case I: Pure Translational Motion

  $v_{B/A} = 0 \qquad w = 0$ Therefore, Equation (5) reduce to $v_B = v_A$

- Case II: General Motion without Sliding on Rotating Link

  $v_{B/A} = 0$ Therefore, Equation (5) reduce to $v_B = v_A + w \times r_{B/A}$

- Case III: General Motion with Sliding on rotating Link

$$v_B = v_A + v_{B/A} + w \times r_{B/A}$$

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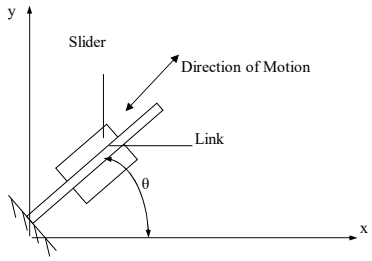
Velocity Analysis using the Vector Method Cases

- Case IV: General Motion with Sliding on Fixed Link (also applicable to cases I, II & III if there is sliding)
  - Consider a body sliding on a link that is inclined at θ to the horizontal in x-y plane.
  - If the magnitude of the sliding velocity is V, then

    $v_{slider} = V(\cos\theta i + \sin\theta j)$
  - $V_{slider} = V_A$ if the link is fixed
  - $V_{slider} = V_{B/A}$ if the link is rotating

    For horizontal motion, $V_{slider} = Vi$

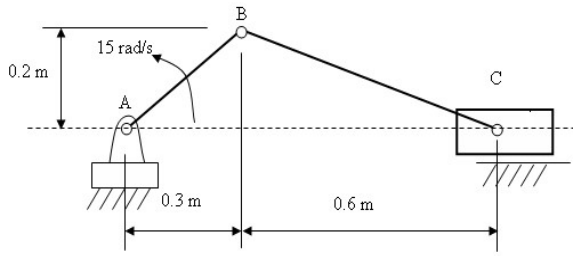
    For vertical motion, $V_{slider} = Vj$



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Example 3.11

Crank AB of crank-slider mechanism shown is rotating at 15 rad/s in counter-clockwise direction. Determine the velocity of slider C and angular velocity of connecting rod BC.



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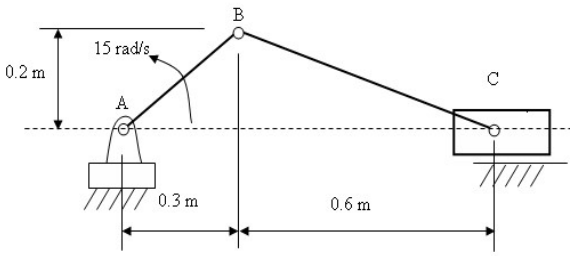
Solution

From A to B

$$v_B = v_A + w \times r_{B/A}$$

$$v_B = v_A + wk \times r_{B/A} = 0 + \begin{bmatrix} i & j & k \\ 0 & 0 & 15 \\ 0.3 & 0.2 & 0 \end{bmatrix}$$

$$v_B = -3i + 4.5j \text{ (m/s)}$$



From B to C

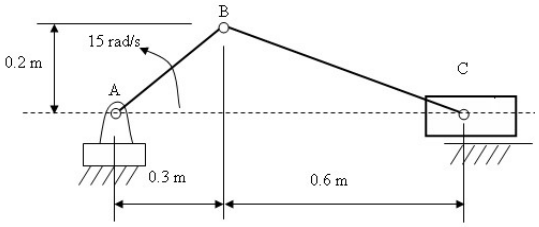
$$v_C = v_B + w_{BC} \times r_{C/B}$$

$$v_C i = (-3i + 4.5j) + \begin{bmatrix} i & j & k \\ 0 & 0 & \omega_{BC} \\ 0.6 & -0.2 & 0 \end{bmatrix}$$

$$v_C = (0.2\omega_{BC} - 3.0)i + (0.6\omega_{BC} + 4.5)j \quad (1)$$

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Solution



Sliding Motion

$$v_C = V_c(\cos\theta i + \sin\theta j) = v_c\left(\cos(0^o)i + \sin(0^o)j\right) = V_C i \quad (2)$$

Equating the i and j components of equation (2) yields

$$v_C = 0.2\omega_{BC} - 3.0$$
$$0 = 0.6\omega_{BC} + 4.5$$

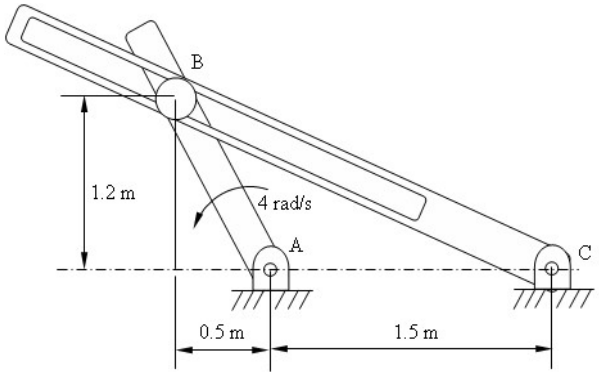
Solving the above two equations simultaneously gives

$$\omega_{BC} = -7.5\, rad/s$$
$$v_C = -4.5 \text{ m/s}$$

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Example 3.12

For the mechanism shown, determine the angular velocity of bar BC and the magnitude of the sliding velocity of the pin at B if crank AB rotates at 4 rad/s in the anticlockwise direction



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Solution

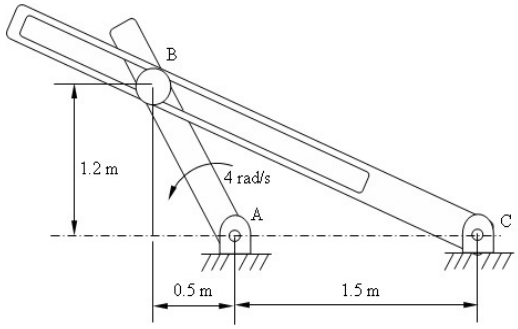
From A to B

$$v_B = v_A + w \times r_{B/A}$$

$$v_A = 0 \qquad \omega = 4\,\text{k} \qquad r_{B/A} = -0.5i + 1.2j \text{ (m)}$$

$$v_B = 0 + 4k \times (-0.5i + 1.2j)$$

$$v_B = -4.8i - 2j \qquad (1)$$



From C to B

$$v_B = v_C + v_{B/C} + w_{CB} \times r_{B/C}$$

$$v_C = 0 \qquad r_{B/C} = -2.0i + 1.2j \text{ (m)}$$

$$v_B = 0 + v_{B/C} + \omega_{CB} k(-2.0i + 1.2j)$$

$$v_B = v_{B/C} - 1.2\omega_{CB} i - 2\omega_{CB} j \qquad (2)$$

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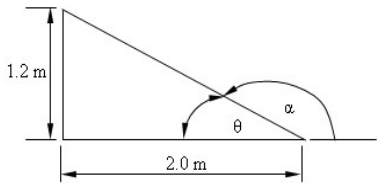
Solution

Let V be the magnitude of the sliding velocity of the pin at B

$$v_{B/C} = V(\cos\theta i + \sin\theta j) \qquad (3)$$

where θ is the angle the sliding direction makes with the horizontal. The angle may be calculated as

$$\theta = 180 - \tan^{-1}\left(\frac{1.2}{2}\right) = 149.04^o$$



$$v_{B/C} = V(\cos 149.03 i + \sin 149.03 j)$$

$$v_{B/C} = V(-0.8575i + 0.5145j)$$

Substituting the above equation into equation (2) yields

$$v_B = V(-0.8575i + 0.5145j) - 1.2\omega_{CB} i - 2\omega_{CB} j$$

$$v_B = -(1.2\omega_{CB} + 0.8575V)i + (-2\omega_{CB} + 0.5145V)j \qquad (4)$$

Equating equations (1) and (4), and equating *i* and *j* components of the above equation gives

$$4.8 = 1.2\omega_{CB} + 0.8575V \qquad (5)$$

$$2 = 2\omega_{CB} - 0.5145V \qquad (6)$$

Solving the equations (5) and (6) simultaneously gives

$$\omega_{CB} = 1.79 \text{ rad/s}$$

$$V = 3.09 \text{ m/s}$$

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From Velocity Analysis

$$r_B = r_A + r_{B/A} \qquad (1)$$

- Where $r_{B/A}$ is the position of point B relative to point A. Let

$$r_{B/A} = xi + yj + zk \qquad (2)$$

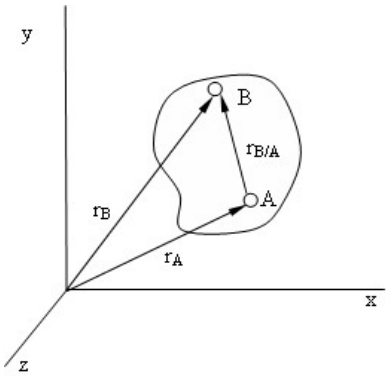
Differentiation Equation (1) with respect to time gives

$$v_B = v_A + \left(\frac{dx}{dt}i + x\frac{di}{dt}\right) + \left(\frac{dy}{dt}j + y\frac{dj}{dt}\right) + \left(\frac{dz}{dt}k + z\frac{dk}{dt}\right)$$

$$v_B = v_A + \left(\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right) + \left(x\frac{di}{dt} + y\frac{dj}{dt} + z\frac{dk}{dt}\right) \qquad (3)$$

Differentiating the unit vectors with respect to time only causes rotation of the vectors. Thus,

$$\frac{di}{dt} = w \times i \qquad \frac{dj}{dt} = w \times j \qquad \frac{dk}{dt} = w \times k \qquad (4)$$



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Velocity Analysis using the Vector Method

Substituting the above equation (4) into equation (3), we have

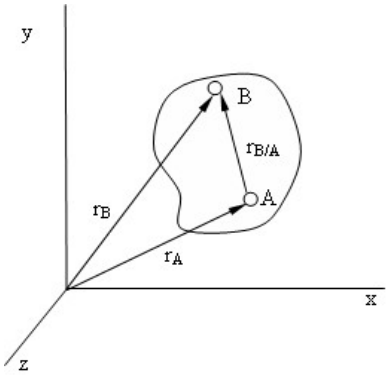
$$v_B = v_A + \left(\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right) + \left[x(\omega \times i) + y(\omega \times j) + z(\omega \times k)\right]$$

$$v_B = v_A + \left(\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right) + \omega \times \left[xi + yj + zk\right]$$

$$v_B = v_A + v_{B/A} + w \times r_{B/A} \qquad (5)$$

where $v_{B/A} = \left(\frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k\right)$

v_A and v_B are respectively velocities of point A and B relative to the reference frame, and $V_{B/A}$ is the sliding velocity of point B relative to point A



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Acceleration Equation

$a_B = a_A + \frac{d}{dt} v_{B/A} + \frac{d}{dt}(w \times r_{B/A})$ ----------(1)

$$\frac{d}{dt} v_{B/A} = \frac{d}{dt}\left(\frac{dx}{dt} i + \frac{dy}{dt} j + \frac{dz}{dt} k\right)$$

$$\frac{d}{dt} v_{B/A} = \left(\frac{d^2x}{dt^2} i + \frac{d^2y}{dt^2} j + \frac{d^2z}{dt^2} k\right) + \omega \times \left[\frac{dx}{dt} i + \frac{dy}{dt} j + \frac{dz}{dt} k\right]$$

$$\frac{d}{dt} v_{B/A} = a_{B/A} + w \times V_{B/A}$$ ----------(2)

$$\frac{d}{dt}\left(w \times r_{B/A}\right) = \frac{d}{dt} w \times r_{B/A} + w \times \frac{d}{dt} r_{B/A}$$

$$\frac{d}{dt}\left(w \times r_{B/A}\right) = \alpha \times r_{B/A} + w \times \left(v_{B/A} + w \times r_{B/A}\right)$$ ----------(3)

Substituting (2) and (3) into (1) and simplifying, we have

$$a_B = a_A + a_{B/A} + 2\omega \times v_{B/A} + \alpha \times r_{B/A} + \omega \times (\omega \times r_{B/A})$$ Spatial (3D) motion

$$a_B = a_A + a_{B/A} + 2\omega \times v_{B/A} + \alpha \times r_{B/A} - \omega^2 r_{B/A}$$ Planer (2D) motion

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Velocity Analysis using the Vector Method Cases

- Case I: Pure Translational Motion

  $a_{B/A} = 0, \alpha = 0$ Therefore, Equation (5) reduce to $a_B = a_A$

- Case II: General Motion without Sliding on Rotating Link

  $a_{B/A} = 0$ Therefore, Equation (5) reduce to $a_B = a_A + \alpha \times r_{B/A} - \omega^2 r_{B/A}$

- Case III: General Motion with Sliding on rotating Link

$$a_B = a_A + a_{B/A} + 2\omega \times v_{B/A} + \alpha \times r_{B/A} - \omega^2 r_{B/A}$$

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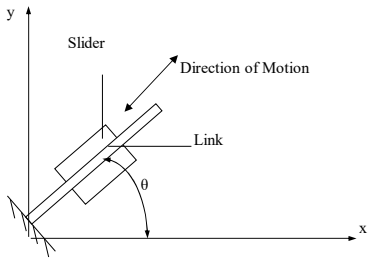
Acceleration Analysis using the Vector Method Cases

- Case IV: General Motion with Sliding on Fixed Link (also applicable to cases I, II & III if there is sliding)
  - Consider a body sliding on a link that is inclined at θ to the horizontal in x-y plane.
  - If the magnitude of the sliding velocity is A, then

    $A_{slider} = A(\cos\theta \boldsymbol{i} + \sin\theta \boldsymbol{j})$
  - $a_{slider} = a_A$ if the link is fixed
  - $a_{slider} = a_{B/A}$ if the link is rotating

    For horizontal motion, $A_{slider} = A\boldsymbol{i}$

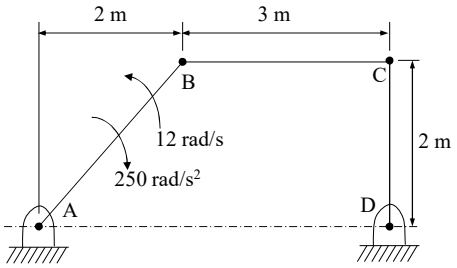
    For vertical motion, $A_{slider} = A\boldsymbol{j}$



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Example 4.1

Link AB shown has an anti-clockwise angular velocity of 12 rad/s and a clockwise angular acceleration of 250 rad/s^2. Determine the angular accelerations of links BC and CD.



Solution

Since the angular acceleration formula has velocity components, first the velocities of the links BC and CD are determined. Then, the determined velocities are used to determine accelerations of the links.

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Velocity Analysis

Then, the determined velocities are used to determine accelerations of the links. The velocity of point B

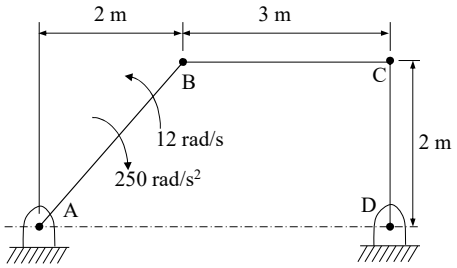
$$v_B = v_A + \omega_{AB} \times r_{B/A} \qquad v_B = 0 + 12\,k \times (2i + 2j)$$

$$v_B = -24i + 24j \quad \text{(m/s)}$$

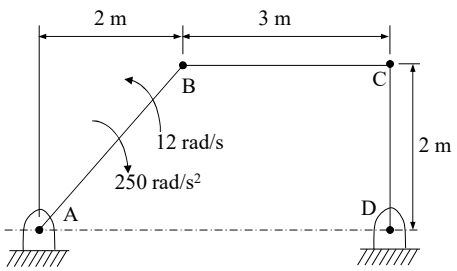
Let ω_{BC} and ω_{CD} the angular velocities of links BC and CD, respectively. The velocity of point C in terms of the velocity of point B is

$$v_C = v_B + \omega_{BC} \times r_{C/B} \qquad v_C = (-24i + 24j) + (\omega_{BC}k) \times (3i)$$

$$v_C = -24i + (3\omega_{BC} + 24)j \quad \text{----------(1)}$$



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Similarly, the velocity of point C in terms of the velocity of point D is

$$v_C = v_D + \omega_{CD} \times r_{C/D}$$

$$v_C = 0 + (\omega_{CD}k) \times (2j) = -2\omega_{CD}i \quad \text{----------(2)}$$

Equating the two expressions for the velocity of point C, i.e. equations (1) and (2), yields

$$-24i + (3\omega_{BC} + 24)j = -2\omega_{CD}i \quad \text{----------(3)}$$

Equation the i and the j components of the above equation and solving for the angular velocities, we have

For i: $-2\omega_{CD} = -24$ $\qquad \omega_{CD} = 12$ rad/s

For j: $(3\omega_{BC} + 24) = 0$ $\qquad \omega_{BC} = -8$ rad/s

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Acceleration Analysis

The acceleration of point B in terms of that of point A is

$$a_B = a_A + \alpha \text{ x } r_{B/A} - \omega^2 r_{B/A}$$

$$a_B = 0 - 250\text{k x}(2\text{i} + 2\text{j}) - (12^2)(2i + 2j)$$

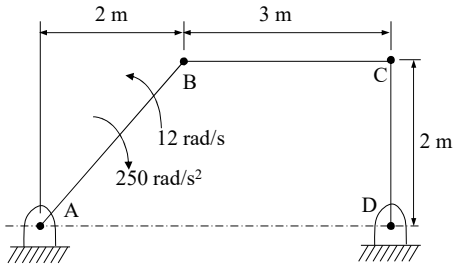
$$a_B = 212\,i - 788\,j$$

Let α_{BC} and α_{CD} the angular acceleration of links BC and CD, respectively. The acceleration of point C in terms of the acceleration of point B is

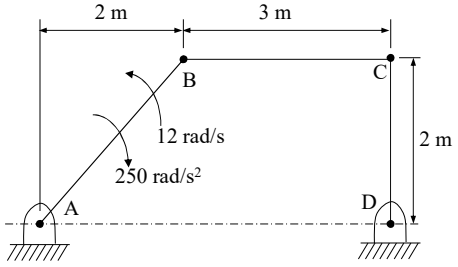
$$a_C = a_B + \alpha_{BC} \text{ x } r_{C/B} - \omega_{BC}^2 r_{C/B}$$

$$a_C = (212i - 788j) + \alpha_{BC}\text{k x } 3\text{i} - (-8)^2(3i)$$

$$a_C = 20i + (3\alpha_{BC} - 788)j \qquad \text{----------(4)}$$



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Similarly, the acceleration of point C in terms of the velocity of point D is

$$a_C = a_D + \alpha_{CD} \text{ x } r_{D/C} - \omega_{CD}^2 r_{C/D}$$

$$a_C = 0 + \alpha_{CD}\text{k x } 2\text{j} - (12)^2(2j)$$

$$a_C = -2\alpha_{CD}\text{i} - 288j \qquad \text{----------(5)}$$

Equating i and j components of Equations (4) and (5) and solving for the angular velocities, we have

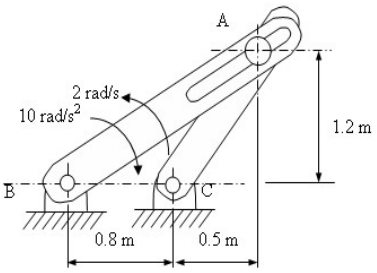
For i: $-2\alpha_{CD} = 20$ $\quad \alpha_{CD} = -10$ rad/s²

For j: $3\alpha_{BC} - 788 = -288$ $\quad \alpha_{BC} = 166.7$ rad/s²

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Example 4.2

For the mechanism shown, determine the angular acceleration of bar AC and acceleration of pin A relative to the slot in bar AB if bar AB has an anticlockwise velocity of 2 rad/s and a clockwise angular acceleration of 10 rad/s^2



Solution

Velocity of point A is given as

$$v_A = v_B + v_{A/B} + \omega_{BA} \text{ x } r_{A/B}$$

$$v_A = 0 + v_{A/B} + 2\text{k x}(1.3i + 1.2j) \quad (1)$$

$$v_A = v_{A/B} + (-2.4i + 2.6j) \quad (2)$$

$$v_{A/B} = |v_{A/B}|(\cos\theta i + \sin\theta j)$$

Where $\theta = \tan^{-1}\left(\frac{1.2}{1.3}\right) = 42.709^o$

Substituting for θ and (2) into (1) we have

$$v_A = (0.7384|v_{A/B}| - 2.4)i + (0.6783|v_{A/B}| + 2.6)j \quad (3)$$

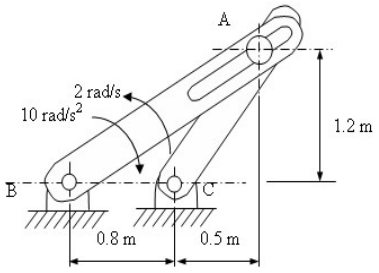
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Example 4.2

Let ω_{CA} and α_{CA} be the angular velocity and angular acceleration of link CA. The velocity of point A from C is

$$v_A = v_C + \omega_{CA} \text{ x } r_{A/C} \qquad v_A = 0 + \begin{bmatrix} i & j & k \\ 0 & 0 & \omega_{CA} \\ 0.5 & 1.2 & 0 \end{bmatrix}$$

$$v_A = -1.2\omega_{CA} i + 0.5\omega_{CA} j \quad (4)$$



Equating the two velocities of A gives

$$-1.2\omega_{CA} i + 0.5\omega_{CA} j = (0.7384 v_{Arel} - 2.4)i + (0.6783 v_{Arel} + 2.6) \quad (5)$$

Equating i and j components of the above equation yields

$$-1.2\omega_{CA} = (0.7384 v_{Arel} - 2.4) \quad (6)$$

$$0.5\omega_{CA} = (0.6783 v_{Arel} + 2.6) \quad (7)$$

Solving the above two equations, we have

$$\omega_{CA} = 2.995 \qquad v_{A/B} = -1.625$$

The velocity and acceleration of A relative B are resolved into i and j components as

$$v_{A/B} = -3.5258(\cos 42.709 i + \cos 42.709 j) = -2.5907i - 2.3915j$$

$$\vec{a}_{A/B} = \alpha_{A/B}(\cos 42.709 i + \cos 42.709 j) = 0.7348\alpha_{A/B} i + 0.6783\alpha_{A/B} j$$

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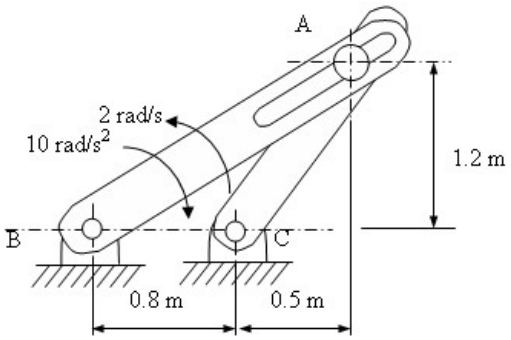
Example 4.2

The acceleration of point A given the velocity of B is

$$a_A = a_B + a_{A/B} + 2\omega_{AB} \times v_{A/B} + \alpha_{BA} \times r_{A/B} - \omega^2 r_{A/B}$$

$$a_A = 0 + 0.7348\alpha_{A/B} i + 0.6783\alpha_{A/B} j + 2\begin{bmatrix} i & j & k \\ 0 & 0 & 2 \\ -2.5907 & -2.3915 & 0 \end{bmatrix} + \begin{bmatrix} i & j & k \\ 0 & 0 & 10 \\ 1.3 & 1.2 & 0 \end{bmatrix} - (10)^2 (1.3i + 1.2j)$$

$$a_A = (0.7348\alpha_{A/B} - 137.2171)i + (0.6783\alpha_{A/B} - 112.1815)j \quad (8)$$



Acceleration of A given the velocity of C is

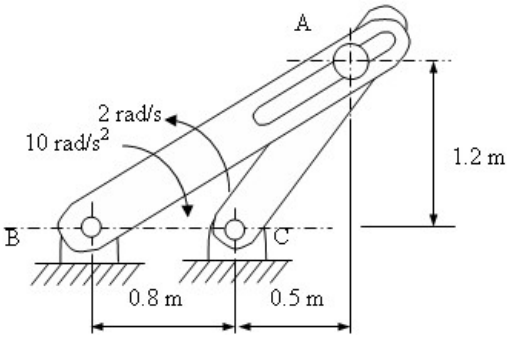
$$a_A = a_C + \alpha_{CA} \times r_{A/C} - \omega^2 r_{A/C}$$

$$a_A = 0 + \begin{bmatrix} i & j & k \\ 0 & 0 & \alpha_{CA} \\ 0.5 & 1.2 & 0 \end{bmatrix} - 2^2 (0.5i + 1.2j)$$

$$a_A = -(1.2\alpha_{CA} + 2)i + (0.5\alpha_{CA} - 1.2j) \quad (9)$$

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Example 4.2



Equating i and j components of the two accelerations of point C yields ie Equations (8) and (9), we have

$$0.7348a_{A/B} - 137.2171 = -1.2\alpha_{CA} + 2 \quad (10)$$

$$0.6783\,a_{A/B} - 112.1815 = 0.5\alpha_{CA} - 1.2 \quad (11)$$

Solving the above two equations simultaneously, we have

$$\alpha_{A/B} = 171.6582 \quad (\text{rad/s}^2)$$

$$a_{A//B} = 10.9017 \quad (\text{m/s}^2)$$

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Example 4.3

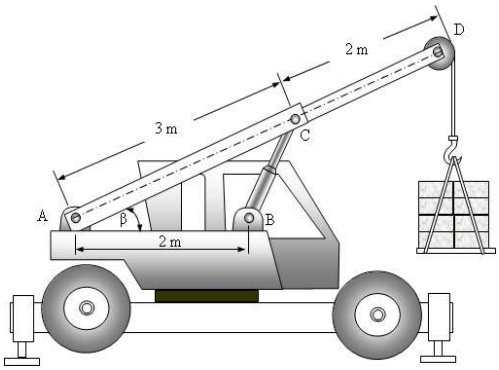
The hydraulic cylinder BC of the crane is extending at a constant rate of 0.2 m/s. When the angle $\beta = 35^o$, Determine

(a) the angular velocity of the crane's boom AD.

(b) the angular velocity of the hydraulic cylinder BC

(c) the angular acceleration of the crane's boom AD.

(d) the angular acceleration of the hydraulic cylinder BC



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